Sonali Ankush Savane

ms25mtech11009@iith.ac.in | sonalisavane11@gmail.com | +91 9130551315 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.69
A S D B Dadawala Jr College Pune	Class XII (Maharashtra State Board)	2015	52.31%
Mahatma Jyotiba Phule Highschool Adambag Pune	Class X (Maharashtra State Board)	2013	80.00%

INTERNSHIP

Safear Defense India | Research Intern - Oct-Nov 2022 (Remote)

Developed 3D structural models in CATIA, improving component & supporting accurate prototype development.

Created 2D technical drawings using SolidWorks & AutoCAD, enabling clear manufacturing & fabrication reference.

Assisted in autopilot hardware testing and debugging, resolving integration issues to improve system functionality

WORK EXPERIENCE

1) Superb Technology Pvt Ltd | Design Engineer - 1 year |

Designed SPM & automation machine components in SolidWorks for accurate and manufacturing-ready designs.

Developed 3D models, assemblies, 2D manufacturing drawings & BOMs, enabling smooth fabrication & assembly.

Developed material handling solutions as per customer requirements for efficient material movemen

2) Indo German Tool Room |Trainee Engineer 1.6 Year |

Trained students in CATIA, AutoCAD, and SolidWorks through practical software-based sessions.

Coordinated placement activities by connecting with companies and organizing recruitment drives.

Managed administrative and admission-related activities to support smooth institute operations.

PROJECT

M.TECH RESEARCH PROJECT | Present |

Modeling of Welding & Characterization of Material

- Conducting welding experiments to study **microstructure, weld penetration and weld quality** under different welding conditions.
- Prepare welded specimens using **metallographic sample preparation and electropolishing** for materials characterization.
- Perform **optical microscopy, SEM and XRD** to investigate microstructure and material characteristics of welded samples.
- Use **RobotStudio** for robotic welding setup and programming to establish and understand welding process conditions.

Solar Cell Prototype - Energy Materials & Devices |

- Applied fundamentals of **energy materials, material properties and device performance.**
- Fabricated a basic **solar-cell prototype** to demonstrate light-to-electricity conversion.
- Tested the prototype under light exposure and measured electrical output.

BTech Research Project:- Design and Fabrication of Nut Segregator Machine |

- Designed Nut Segregator Machine using **SOLIDWORKS & CATIA**.
- Performed **structural analysis in ANSYS** to validate the design.
- Developed **3D models, assemblies, and fabrication drawings**.
- Fabricated and assembled a **working prototype** in the college workshop.
- Sucssefully Tested and demonstrated the **functional prototype** .

Btech Project Simulation of Anti-Lock Braking System (ABS) - MATLAB |

- Simulated an **Anti-Lock Braking System (ABS)** using MATLAB/Simulink.
- Modeled **vehicle braking and wheel dynamics** under different conditions.
- Analyzed **wheel slip, braking force, and stopping performance**.
- Evaluated ABS performance for **improved braking stability and control**.

Diploma Project:- Fabrication of Centrifugal Oil Cleaner |

- Fabricated a **Centrifugal Oil Cleaner** prototype as a college project.
- Assembled the machine components and developed a **working prototype**.
- Tested and demonstrated the **prototype** for oil cleaning application.

IGTR Project:- Design & Analysis of Shock Absorber |

- Designed **Shock Absorber components using CATIA Part Design**.
- Performed **structural analysis using ANSYS**.
- Evaluated **stress and deformation under applied loading conditions**.

IGTR Project- Design and Assembly of Bushed Pin- Flexible Coupling using SOLIDWORKS. |

- Designed and assembled a **Bush Type Flexible Coupling** using SOLIDWORKS.
- Created **3D models and assembly** of individual components.
- Applied appropriate **mates and constraints** for proper assembly.

TECHNICAL SKILLS

- **Software & Tools:** Python, FiPy, ANSYS, COMSOL Multiphysics, OpenFOAM, CrystalMaker, RobotStudio.
- **CAD:** CATIA Part Design, SOLIDWORKS, 3D Modeling & Assembly, GD&T.
- **Materials & Metallurgy:** Welding Metallurgy, Metallography, XRD Analysis, Material Characterization
- TIG Welding | EDM Wire Cut Machine | Machining | Milling | Forging |

CERTIFICATE

- **Master Certificate Course (MCC) in CAD/CAM** -- IGTR Pune | Apr 2022 - Sep 2022
- **Workshop on MicroSim Suite: Methods, Modeling and Applications** -- C-DAC, Pune | 2025
- **National Level Online Quiz - Engineering Metallurgy** -- AISSMS College of Engineering, Pune | 2020
- **Arduino Quiz Competition** -- SGOI College of Engineering, Belhe | 2020

LEADERSHIP & RESPONSIBILITIES

Placement Coordinator - IIT Hyderabad | June-2026 to Present|

- Coordinate placement activities and communicate with companies regarding recruitment opportunities.
- Support coordination between recruiters, institute stakeholders and students

Publicity Head - Annual Function, D.Y. Patil College of Engineering | FEB-2020|

- Managed publicity and promotional activities for the annual college event.
- Coordinated with team members for smooth event promotion and execution.

Coordinator - MESA, D.Y. Patil College of Engineering | 2019-2022|

- Coordinated departmental technical activities, student programs and events

Event Head - Technical Events, IGTR Pune |Jan 2023-June 2024|

- Coordinated technical events and student participation.
- Supported technical event planning and execution

HOBBIES

- **Playing Chess, Kabbadi | Cooking | Traveling | Music**

LANGUAGE

- **MARATHI | HINDI | ENGLISH**
- **Japanese Language - Basic Learning**